

JEFFREY LOWE

Frederick, MD | (301) 676-3252 | jeffrey.lowe.dev@gmail.com

GitHub: <https://github.com/HuntedCode> | Portfolio: <https://jeffreylowe.dev/>

PROFESSIONAL PROFILE

Python developer with hands-on experience designing and maintaining production-scale Django applications, complex data models, and asynchronous processing systems. Demonstrated expertise in backend architecture, API integration, and performance optimization. Known for building scalable systems, solving complex technical problems, and delivering well-documented, production-ready solutions.

TECHNICAL SKILLS

Languages	Python, SQL, JavaScript
Frameworks	Django, FastAPI, discord.py
Databases	PostgreSQL, Redis
Backend & Systems	REST APIs, background processing, async workflows, ETL pipelines
Tools & Platforms	Git, Render, Stripe, PayPal
Concepts	Data modeling, ORM optimization, system architecture, API integration, performance tuning

SELECT PROJECTS

PlatPursuit – Production Django Application

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Production-scale Python web application with 40+ integrated systems, complex relational data models, and real-time data pipelines.

- Built and maintain a Django-based platform (PostgreSQL + Redis) with 49+ documented systems
- Authored 30+ Django models and 50+ class-based views, along with services, serializers, template tags, and management commands
- Designed a normalized relational schema with complex many-to-many relationships, polymorphic data structures, and cascade strategies
- Optimized database performance using advanced Django ORM techniques (select related, prefetch related, Subquery, defer, annotations)
- Diagnosed and resolved a production memory issue caused by inefficient query joins, improving system performance and stability

- Built robust data migration logic (Concept.absorb) to safely merge records across 15+ related models

Background Processing & Async Systems

- Designed and implemented a custom Python worker (TokenKeeper) for API credential management, avoiding heavier frameworks like Celery
- Built a Redis-backed deferred notification system with event consolidation logic to prevent duplication
- Developed cron-based job pipelines for scheduled sync and refresh operations with DST-safe execution windows
- Implemented batching and signal suppression techniques for high-throughput data updates

API Integration & Data Pipelines

- Built a multi-strategy IGDB integration pipeline with retry handling, validation logic, and fallback strategies
- Implemented PlayStation Network sync with token refresh, rate limiting, and system health validation
- Developed ETL pipelines for ingesting and processing external data sources (game releases and trophy systems)
- Integrated Stripe and PayPal payment processing on the backend

Multi-Service Python Architecture

- Built PlatBot using discord.py and FastAPI to extend application functionality via a separate Python service
- Designed API-first architecture enabling shared data models and cross-service communication
- Applied asynchronous programming patterns (asyncio) for efficient event handling

Engineering Process & Documentation

- Authored 49+ technical documents covering system architecture, operational processes, and design standards
- Follow a structured development workflow emphasizing planning, code quality, and documentation alignment

- Maintain disciplined version control practices with integrated documentation updates

PROFESSIONAL EXPERIENCE

Technology Consultant (Freelance) | Frederick, MD | 07/2020 – Present

- Delivered Python-based development solutions including web applications, backend systems, and data pipelines
- Designed scalable architectures and implemented performance optimizations across multiple projects
- Managed full development lifecycle from system design through deployment

Contract Management & IT Support Specialist | FoodPRO Corp | 02/2017 – 07/2020

- Automated business processes using Python and scripting tools, improving efficiency by over 75%
- Streamlined workflows and supported operational improvements across departments
- Assisted leadership in optimizing internal systems and reporting processes

Wastewater Operator (Level 2) | Frederick County Government | 09/2023 – Present

- Operate and troubleshoot SCADA-controlled systems in a complex technical environment
- Monitor real-time system data and adjust operations to maintain performance and stability
- Train new operators and support process improvements

ADDITIONAL EXPERIENCE

- **Automotive Sales Consultant** — Rocket Auto (2021–2022)
- **Assistant Manager** — Black Hog BBQ & Bar (2015–2017)

EDUCATION

High School Diploma — Urbana High School, Ijamsville, MD